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Table of Contents / 本期目录

Article	Title and Author	Page
NN-2026-001	小苏打浓度与煮制时间对意大利面拉面化效果的影响 Effects of Baking Soda Concentration and Cooking Time on the Ramen-Style Transformation of Spaghetti 留子也要好好吃面	3
NN-2026-002	五种商品意大利面在碱水煮制下的尺寸变化、时间进程与中式面条适配性 Dimensional Changes, Time Course, and Chinese-Noodle Applications of Five Commercial Pasta Types 留子也要好好吃面	7

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小苏打浓度与煮制时间对意大利面拉面化效果的影响：一项 3 × 3 探索性厨房实验

Effects of Baking Soda Concentration and Cooking Time on the Ramen-Style Transformation of Spaghetti: A 3 × 3 Exploratory Kitchen Trial

作者 / Author: 留子也要好好吃面 • Liuzi Ye Yao Haohao Chi Mian

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Figure 1. Finished-dish presentation documented by the author. Commercial products were independently purchased.

摘要与关键词 / Abstract and Keywords

摘要： 本探索性厨房实验考察小苏打浓度与煮制时间对商品干制意大利面“拉面化”感官表现的共同影响。实验采用3 × 3处理矩阵，将2、4、6 g/L三种小苏打浓度与10、12、15分钟三种煮制时间组合，共形成九种处理。作者依据弹性、中心熟度、顺滑度、碱香纯度、苦涩感及与拉面风格面条的整体相似度进行结构化描述，并给予10分制综合评分。结果显示，4 g/L、15分钟处理得分最高（9.5/10），表现为颜色金黄、面体顺滑、弹性较好且碱香适中；4 g/L、12分钟得分8.5/10，保留较明显的偏硬中心。2 g/L处理的碱性特征总体较弱，而6 g/L处理随时间延长出现苦味、涩感、表面黏口及肥皂样气味。该结果支持将4 g/L、15分钟作为家庭厨房试制的实用起点，但由于研究为单人评价、未报告独立重复且缺少pH、仪器质构和烹煮损失测量，结论应理解为透明的探索性观察，而非统计学推广结果。

Abstract: This exploratory kitchen trial examined the combined effects of baking-soda concentration and cooking time on the ramen-style transformation of commercial dried spaghetti. A 3 × 3 treatment matrix combined three concentrations, 2, 4, and 6 g/L, with cooking times of 10, 12, and 15 minutes, producing nine treatments. The author documented elasticity, center doneness, surface smoothness, alkaline aroma, bitterness or astringency, and overall similarity to ramen-style noodles, then assigned a descriptive overall score on a 10-point scale. The highest score, 9.5/10, was assigned to 4 g/L for 15 minutes. This treatment was described as golden, smooth, springy, and appropriately alkaline without obvious bitterness. The 4 g/L, 12-minute condition scored 8.5/10 and retained a firmer center. Treatments at 2 g/L produced limited alkaline character, whereas prolonged cooking at 6 g/L produced bitterness, astringency, surface stickiness, and a soap-like odor. The findings support 4 g/L for 15 minutes as a practical starting condition for home use, but the results remain exploratory because the work involved one evaluator, undocumented replication, and no pH, instrumental texture, or cooking-loss measurements.

关键词 / Keywords:
小苏打；意大利面；碱水面；拉面替代；感官评价；厨房实验 / baking soda; spaghetti; alkaline noodles; ramen substitute; sensory evaluation; culinary methods

1. 研究背景 / Background

在部分地区，新鲜碱水面或成品拉面的获取并不稳定，而干制意大利面具有价格可控、规格稳定和储存方便等特点。小苏打进入煮面水后会提高体系碱性，可能影响面条颜色、气味、表面结构和面芯熟化速度。网络烹饪经验常建议“加一点小苏打”来制造拉面感，但浓度和时间通常缺乏系统比较。

Fresh alkaline noodles are not consistently available in every market, whereas dried spaghetti is widely distributed, shelf-stable, and manufactured to a relatively consistent specification. Adding baking soda raises the alkalinity of the cooking medium and may alter noodle color, aroma, surface properties, and center softening. Informal recipes often recommend adding an unspecified amount of baking soda, but concentration and cooking time are rarely compared in a structured matrix.

本研究的目标是：在家庭厨房条件下，以统一水量建立浓度与时间的交叉比较；记录各处理的主要感官差异；识别最有希望的实用组合；并如实说明证据边界。

The objectives were to compare concentration and time under a common water volume, document the sensory differences among treatments, identify a practical candidate condition, and state the limits of the evidence transparently.

2. 材料与方法 / Materials and Methods

材料。 实验使用商品干制硬质小麦意大利面、小苏打和饮用水。包装标示常规煮制时间约10-11分钟。原始图片记录了4 g小苏打称量，但没有记录每组干面质量。

Materials. Commercial dried durum-wheat spaghetti, baking soda, and water were used. The package indicated a conventional cooking time of approximately 10-11 minutes. A 4 g baking-soda measurement was photographed, but dry pasta mass per treatment was not recorded.



Figure 2. Commercial spaghetti and measurement of 4 g baking soda.

实验设计。 每组使用1 L水。小苏打浓度设置为2、4、6 g/L，煮制时间设置为10、12、15分钟，共九组。实验在家庭炉具上完成，原始记录未提供独立重复次数或随机化顺序。

Design. Each treatment used 1 L water. Concentrations were 2, 4, and 6 g/L, and cooking times were 10, 12, and 15 minutes. The trial was conducted on a household stove. Independent replication and randomized treatment order were not documented.

评价方法。 作者根据面条弹性、表面顺滑度、中心硬度、碱香、苦涩或肥皂样异味，以及与拉面风格面条的整体相似度进行结构化描述，并给予综合评分。评分属于单人主观判断，并非训练型感官小组或盲评。

Evaluation. The author recorded springiness, surface smoothness, center firmness, alkaline aroma, bitterness or soap-like odor, and overall ramen-style similarity. The score represents one evaluator’ s structured judgment rather than a blinded or trained sensory panel.

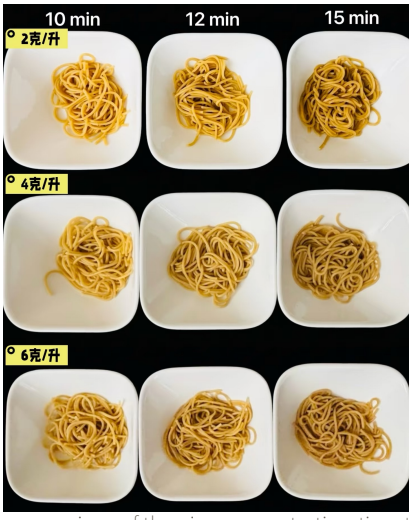


Figure 3. Visual comparison of the nine concentration-time treatments.

3. 结果 / Results

Baking soda	10 min	12 min	15 min
2 g/L	5.0	6.0	6.5
4 g/L	7.0	8.5	9.5
6 g/L	6.0	5.0	3.5

2 g/L组随时间延长仅呈温和改善：10分钟仍接近普通意大利面，12分钟碱香不足，15分钟虽熟透但偏软。4 g/L组呈持续上升趋势：10分钟外弹内硬，12分钟更接近偏硬拉面，15分钟获得最佳综合表现。6 g/L组则随时间延长持续下降：10分钟已出现偏强碱味，12分钟苦涩和黏口，15分钟出现明显肥皂样气味和苦味。

At 2 g/L, scores improved only modestly with time: the 10-minute sample remained spaghetti-like, the 12-minute sample had weak alkaline character, and the 15-minute sample was fully cooked but soft. At 4 g/L, scores improved across time: the 10-minute treatment had an elastic exterior but firm center, 12 minutes approached a firmer ramen style, and 15 minutes produced the best overall result. At 6 g/L, scores declined with time as bitterness, astringency, stickiness, and soap-like odor intensified.



Figure 4. Original score matrix and treatment descriptions.

4. 实用方法 / Practical Procedure

推荐起始方法为：向1 L水中加入4 g小苏打并煮沸；加入意大利面后继续煮15分钟；使用容量较大的锅以应对泡沫；捞出后用于汤面或拌面。偏好更硬面芯时可将时间缩短至约12分钟。该方法应被视为起始条件，实际时间仍会受到面条直径、炉具功率、海拔和品牌配方影响。

A practical starting method is to add 4 g baking soda to 1 L water, bring the solution to a boil, add spaghetti, and cook for 15 minutes. A larger pot helps manage foaming. The noodles may then be used in soup or mixed-noodle preparations. A 12-minute endpoint may suit a firmer preference. Required time can vary with strand diameter, stove output, altitude, and product formulation.

5. 讨论 / Discussion

实验结果最重要的模式并非“碱越多越好”，而是中等浓度与充分熟化之间的平衡。2 g/L可能不足以产生明显的碱水面特征；6 g/L则容易在延长煮制时积累令人不悦的苦涩和肥皂样风味。4 g/L提供了较大的可操作窗口，在12至15分钟之间可以通过时间调节面芯硬度。

The central pattern was not that more alkali produced a better noodle. Instead, a middle concentration had to be balanced with sufficient center softening. At 2 g/L, alkaline character was limited. At 6 g/L, prolonged cooking amplified undesirable flavor and surface properties. The 4 g/L condition created a usable window in which cooking time could be adjusted between approximately 12 and 15 minutes to change center firmness.

照片中的颜色随浓度和时间总体加深，但该变化不能被当作仪器色差结果。相机曝光、白平衡、拍摄时间和样品表面水分均可能影响视觉判断。

Color appeared to darken with concentration and time, but the photographs are not instrumental color measurements. Exposure, white balance, time after cooking, and surface moisture could all influence appearance.

6. 局限性 / Limitations

本研究未报告每组干面质量、独立重复次数、随机化与盲评流程，也没有独立评审人数。10分制综合评分未经统计验证。未测量pH、含水率、烹煮损失、断裂力、拉伸强度或仪器色差。普通水煮意大利面虽被提及为概念性对照，但未提供独立评分。因此，本研究应被定位为可供复现实验设计参考的探索性厨房报告。

Dry pasta mass, independent replication, randomized order, blinding, and the number of assessors were not documented. The 10-point score was not statistically validated. pH, moisture, cooking loss, breaking force, tensile properties, and instrumental color were not measured. A conventional-water control was mentioned but not scored separately. The study should therefore be interpreted as an exploratory culinary report that can inform a more rigorous replicated experiment.

7. 结论 / Conclusion

在本次九组探索性比较中，4 g/L小苏打、15分钟处理获得最高评分，4 g/L、12分钟提供更硬的替代选择。高浓度和长时间组合表现最差。后续研究应增加重复实验、独立感官评价、pH和仪器质构测量，并记录干面质量与烹煮损失。

Among the nine treatments, 4 g/L for 15 minutes received the highest score, while 4 g/L for 12 minutes provided a firmer alternative. High concentration combined with prolonged cooking performed worst. Future work should add replication, independent sensory assessment, pH and instrumental texture measurements, dry-mass standardization, and cooking-loss analysis.

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五种商品意大利面在碱水煮制下的尺寸变化、时间进程与中式面条适配性：一项探索性厨房比较

Dimensional Changes, Time Course, and Chinese-Noodle Applications of Five Commercial Pasta Types Cooked in Alkaline Water: An Exploratory Kitchen Comparison

作者 / Author: 留子也要好好吃面 • Liuzi Ye Yao Haohao Chi Mian

Article type: Exploratory Culinary Research Report License: CC BY 4.0



Figure 1. Finished-dish presentation documented by the author. Commercial products were independently purchased.

摘要与关键词 / Abstract and Keywords

摘要： 本探索性厨房比较在统一碱水条件下考察五种商品意大利面的尺寸变化、建议煮制时间和中式面条应用潜力。Angel Hair、Thin Spaghetti、Spaghetti、Linguine和Fettuccine均在1 L沸水加入4 g小苏打的条件下煮制，并在作者判断的“刚好弹牙”端点记录时间。所报告的生面/熟面尺寸分别为1.2/1.8、1.4/2.1、1.8/2.8、3.1/4.6和4.6/6.8 mm；对应建议时间为5、9、15、17和18分钟。五种产品的平均尺寸膨胀倍数为1.50。Angel Hair的3至8分钟时间序列显示，测量直径由1.5增加至1.9 mm，但推荐感官端点为5分钟，说明最大膨胀并不等同于最佳口感。名义生面尺寸与建议煮制时间呈描述性正相关（Pearson r = 0.82），但样本仅包含五种产品，且圆面使用直径、扁面使用宽度，因此该相关不能用于推断一般规律。结果可为家庭烹饪者选择面型和时间提供初步参考，同时也提出了标准化测量与重复实验的需要。

Abstract: This exploratory kitchen comparison examined dimensional change, recommended cooking time, and potential Chinese-noodle applications for five commercial pasta formats under a standardized alkaline cooking condition. Angel Hair, Thin Spaghetti, Spaghetti, Linguine, and Fettuccine were cooked in 1 L boiling water containing 4 g baking soda, and cooking time was recorded at the author-defined al dente endpoint. Reported dry/cooked dimensions were 1.2/1.8, 1.4/2.1, 1.8/2.8, 3.1/4.6, and 4.6/6.8 mm, with recommended times of 5, 9, 15, 17, and 18 minutes. Mean dimensional expansion was 1.50. An Angel Hair time series from 3 to 8 minutes increased from 1.5 to 1.9 mm, although 5 minutes was selected as the preferred sensory endpoint, indicating that maximum swelling did not equal optimal texture. Nominal dry size and recommended time showed a descriptive positive association (Pearson r = 0.82). However, the dataset included only five products and mixed diameter measurements for round pasta with width measurements for flat pasta. The findings offer an initial practical reference while highlighting the need for replication and standardized measurement.

关键词 / Keywords:
碱水煮制；卡尺测量；尺寸膨胀；意大利面规格；中式面条替代 / alkaline cooking; caliper measurement; dimensional expansion; pasta format; Chinese noodle substitute

1. 研究背景 / Background

不同意大利面规格在粗细、截面形状和包装建议时间上差异明显。若统一采用碱水煮制，家庭烹饪者需要知道细面是否更快达到理想端点、扁面与圆面的尺寸变化是否接近，以及不同规格更适合哪些中式面条菜式。本研究在第一篇4 g/L小苏打条件的基础上，将比较扩展到五种常见规格。

Commercial pasta formats differ in strand thickness, cross-sectional geometry, and conventional cooking instructions. Under a common alkaline cooking condition, a home cook needs to know whether finer strands reach a preferred endpoint sooner, whether flat and round pasta show similar relative dimensional change, and which formats may suit particular Chinese-style dishes. Building on the 4 g/L condition identified in the first report, this study compared five common formats.

2. 材料与方法 / Materials and Methods

样品。 比较对象为Angel Hair、Thin Spaghetti、Spaghetti、Linguine和Fettuccine。原始图片记录了商品包装和名义规格。所有产品由作者自行购买。

Samples. Angel Hair, Thin Spaghetti, Spaghetti, Linguine, and Fettuccine were compared. Product packaging and nominal dimensions were documented photographically. All products were independently purchased by the author.



Figure 2. Commercial pasta products included in the comparison.

统一条件。 每种面条在1 L沸水加入4 g小苏打的条件下煮制。达到作者判断的“刚好弹牙”状态时记录时间。该端点为感官判断，并非仪器终点。

Standardized condition. Each pasta type was cooked in 1 L boiling water containing 4 g baking soda. Time was recorded at the author-defined al dente endpoint. This endpoint was sensory rather than instrumental.

尺寸测量。 使用数显卡尺记录尺寸。三种圆形面记录直径；Linguine和Fettuccine记录宽度。由于熟面柔软且可压缩，卡尺夹持力度可能影响读数。原始材料未说明每种面条测量了多少根、是否进行了重复测量或卡尺是否校准。

Dimensional measurement. A digital caliper was used. Diameter was recorded for the three round formats, whereas width was recorded for Linguine and Fettuccine. Cooked pasta is soft and compressible, so caliper pressure could affect readings. Strand count, repeated measurement, and calibration were not documented.

派生指标。 尺寸膨胀倍数按熟面尺寸除以生面尺寸计算。五种产品的平均倍数为1.50。生面名义尺寸与建议时间的Pearson相关系数为0.82，仅用于描述。

Derived measures. Expansion ratio was calculated as cooked dimension divided by dry dimension. The mean ratio was 1.50.

Pearson correlation between nominal dry dimension and recommended time was 0.82 and is reported descriptively only.

3. 结果 / Results

Type	Dry	Cooked	Ratio	Time
Angel Hair	1.2	1.8	1.50x	5 min
Thin Spag.	1.4	2.1	1.50x	9 min
Spaghetti	1.8	2.8	1.56x	15 min
Linguine*	3.1	4.6	1.48x	17 min
Fettuccine*	4.6	6.8	1.48x	18 min

* Linguine and Fettuccine values are widths; round-pasta values are diameters.

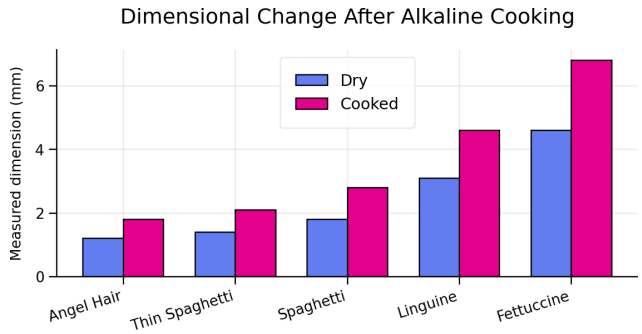


Figure 3. Reported dry and cooked dimensions.

相对膨胀倍数集中在1.48至1.56之间，说明在本次测量中不同规格的相对尺寸增长较为接近；然而绝对增长量随规格增大而增加。标准Spaghetti报告的比例最高（1.56），两种扁面约为1.48。

Relative expansion ratios clustered from 1.48 to 1.56, suggesting similar proportional dimensional growth within this small dataset, while absolute growth increased with pasta size. Standard Spaghetti had the highest reported ratio, 1.56, whereas the two flat formats were approximately 1.48.

建议煮制时间从Angel Hair的5分钟增加到Fettuccine的18分钟。名义尺寸与建议时间呈正相关，但标准Spaghetti位于简单线性趋势之上，而Fettuccine位于趋势之下，提示截面形状、配方和感官端点同样重要。

Recommended time increased from 5 minutes for Angel Hair to 18 minutes for Fettuccine. Nominal size and recommended time were positively associated, but standard Spaghetti sat above a simple linear trend and Fettuccine below it, indicating that geometry, formulation, and sensory endpoint also mattered.

4. Angel Hair时间进程 / Angel Hair Time Course

Angel Hair在3、4、5、6、7、8分钟的读数分别为1.5、1.6、1.7、1.8、1.8和1.9 mm。3分钟时仍可见硬芯；5分钟被选为推荐端点。6分钟以后尺寸继续增加，但原始观察指出面条更容易偏软。

Angel Hair readings at 3, 4, 5, 6, 7, and 8 minutes were 1.5, 1.6, 1.7, 1.8, 1.8, and 1.9 mm. A firm center remained visible at 3 minutes, and 5 minutes was selected as the recommended endpoint. Measured size continued to increase after 5 minutes, but the source observations described progressive softening beyond approximately 6 minutes.

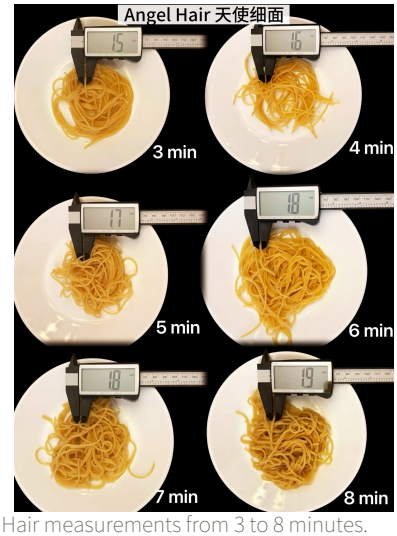


Figure 4. Angel Hair measurements from 3 to 8 minutes.

该序列说明，尺寸膨胀是持续的物理变化，而“最佳口感”是由面芯、表面软化和弹性共同决定的感官端点，两者不应混为一谈。

The series shows that dimensional swelling is a continuing physical change, whereas preferred texture is a sensory endpoint determined jointly by center doneness, surface softening, and elasticity. The two concepts should not be treated as equivalent.

5. 中式面条应用 / Chinese-Noodle Applications

Angel Hair可作为细面、凉面或馄饨面风格的起点，但吸水快、容错窗口短。Thin Spaghetti在细度和耐煮性之间较平衡，可用于阳春面或细拌面。标准Spaghetti的适用范围最广，可用于汤面、热干面风格或炒面实验。Linguine的扁平表面有利于酱汁附着，适合拌面和炒面。Fettuccine宽度较大，可用于烩面或宽面风格。以上建议描述的是烹饪适配性，不表示原料、加工工艺或物理结构与对应中式面条完全相同。

Angel Hair offers a starting point for fine-noodle, cold-noodle, or wonton-noodle-style dishes, but its rapid hydration creates a narrow timing window. Thin Spaghetti balances fineness and cooking tolerance. Standard Spaghetti appears the most versatile for soup, mixed, or stir-fried applications. Linguine’s flat surface supports sauce adhesion, while Fettuccine may suit broad-noodle-style dishes. These suggestions concern culinary use and do not imply equivalence in ingredients, processing, or structure.



Figure 5. Original five-format results and application suggestions.

6. 讨论 / Discussion

五种产品都接近1.5倍的尺寸膨胀，为家庭烹饪提供了一个易记的经验值，但该数字不能替代对单一产品的实际观察。尤其是扁面的宽度变化与圆面的直径变化并非同一物理量，不能据此推断截面积、体积或吸水率。

An approximate 1.5-fold dimensional change offers a memorable practical observation, but it should not replace direct observation of a specific product. Width change in flat pasta and diameter change in round pasta are not the same physical quantity and cannot be used to infer cross-sectional area, volume, or water uptake.

Pearson r = 0.82看似较高，但仅基于五个异质产品。没有重复测量、误差范围或预先设定的统计分析，因此不应报告显著性或建立预测公式。该图的价值主要在于显示“规格通常越大，煮制时间越长”的方向性趋势。

Although Pearson r = 0.82 appears high, it is based on only five heterogeneous products without replication, uncertainty estimates, or a prespecified statistical analysis. Significance testing and prediction equations would therefore be inappropriate. The graph is useful primarily for showing the directional tendency that larger formats generally required longer cooking.

7. 局限性 / Limitations

本研究未说明每种产品测量的面条根数、取样位置、卡尺夹持力度、重复次数和设备校准。包装名义尺寸可能来自制造商而非本次实验直接测量。感官端点由一位作者确定，且不同品牌的蛋白质含量、挤压工艺和干燥条件可能影响结果。全麦意大利面和泡打粉仅有描述性尝试，不纳入定量比较。

The strand count, sampling location, caliper pressure, number of repeats, and instrument calibration were not documented. Nominal dry dimensions may have come from packaging rather than direct measurement. One author selected the sensory endpoint, and differences in protein content, extrusion, and drying may influence results. Descriptive trials involving whole-wheat pasta and baking powder were not included in the quantitative comparison.

8. 结论 / Conclusion

在统一4 g/L小苏打条件下，五种规格的建议时间为5至18分钟，平均尺寸膨胀约1.50倍。较大规格一般需要更长时间，但几何形状和感官端点同样重要。后续研究应使用重复测量、固定卡尺压力、记录品牌与配方，并结合烹煮损失和仪器质构。

Under a standardized 4 g/L baking-soda condition, recommended cooking times ranged from 5 to 18 minutes and mean dimensional expansion was approximately 1.50. Larger formats generally required longer cooking, while geometry and sensory endpoint also mattered. Future work should use repeated measurements, controlled caliper pressure, product-composition records, cooking-loss analysis, and instrumental texture testing.

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